1 Here are the first five terms of a sequence.

2 4 7 11 16

+2 +3 +4 +5

Write down the next two terms in the sequence.

16 + 6 = 22

22 + 7 = 29

22 , 29

(Total for Question 1 is 2 marks)

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2 The first term in a sequence is 3.
The term to term rule is add 5.

3 8 13 18 23 ...

Is 97 a term in the sequence?
Give a reason for your answer.

No, all terms in the sequence end in 3 or 8

(Total for Question 2 is 2 mark)

---

3 Here are the first five terms of a Fibonacci sequence

1 2 3 5 8

Write down the next two terms in the sequence.

5 + 8 = 13

8 + 13 = 21

13 , 21

(Total for Question 3 is 2 marks)

---

4 The nth term of a sequence is 4n + 3

(a) Find the first two terms of this sequence.

4(1) + 3 = 7

4(2) + 3 = 11

7 , 11

(b) Is 35 a term in this sequence.
You must show how you get your answer.

4n + 3 = 35

4n = 32

n = 8

Yes, 35 is the 8th term in the sequence

(Total for Question 4 is 2 marks)

5 The nth term of a sequence is $n^2 + 1$

$(1)^2 + 1 = 2$

$(2)^2 + 1 = 5$

(a) Find the first two terms of this sequence.

2 , 5

(1)

(b) Is 35 a term in this sequence.
You must show how you get your answer.

$n^2 + 1 = 35$

$n^2 = 34$

$n = \sqrt{34}$ [not a whole no.]

No, 35 is not one more than a square number

(1)

(Total for Question 5 is 2 marks)

---

6 Here are the first 5 terms of a sequence.

17 14 11 8 5

(a) Find the next term of this sequence.

2

(1)

The nth term of a different sequence is $10n^2 + 5$

(b) Work out the 5th term of this sequence.

$10(5)^2 + 5$

$10(25) + 5$

$250 + 5$

255

(1)

(Total for Question 6 is 2 marks)

---

7 Here are the first four terms of a sequence.

7 13 19 25

(a) Write down the next term in the sequence.

31

(1)

(b) Explain how you got your answer

added 6 onto the previous term

(1)

(Total for Question 7 is 2 marks)

8 Here are the first four terms of a number sequence.

2 3 5 9

The rule to continue the sequence is
multiply the previous term by 2 and then subtract 1

Work out the 5th term of this sequence.

$9 \times 2 = 18$
$18 - 1 = 17$

17

(Total for Question 8 is 1 mark)

---

9 Here are the first 5 terms of a Fibonacci sequence.

2 2 4 6 10

Find the 8th term of this sequence.

$6 + 10 = 16$
$10 + 16 = 26$
$16 + 26 = 42$

42

(Total for Question 9 is 2 marks)

---

10 The nth term of a sequence is $n^2 + 3$

(a) Find the first three terms of this sequence.

$(1)^2 + 3 = 4$
$(2)^2 + 3 = 7$
$(3)^2 + 3 = 12$

4, 7, 12

(2)

(b) Find the 10th term in this sequence.

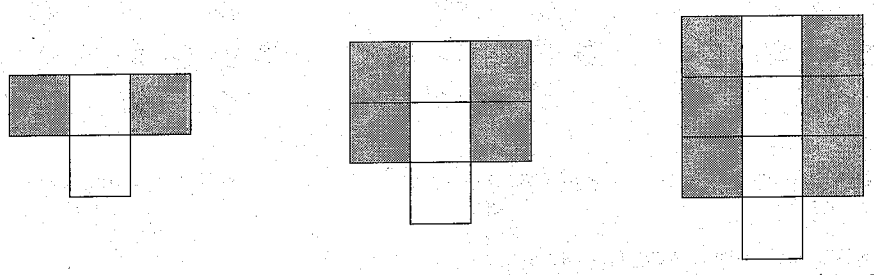
$(10)^2 + 3$
$100 + 3$

103

(1)

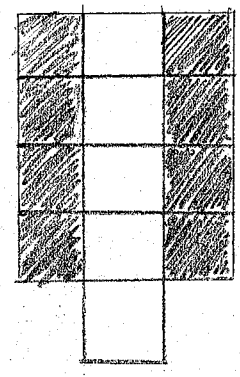
(Total for Question 10 is 3 marks)

11 Here is a sequence of patterns made from white tiles and grey tiles.



pattern number 1 pattern number 2 pattern number 3

(a) In the space below, draw pattern number 4.



(1)

(b) Work out the total number of tiles to make pattern number 7.

4 7 10 13 16 19 22

22

(2)

Kyle says
"There are 4 white tiles in pattern number 3 so there will be 8 white tiles in pattern number 6."

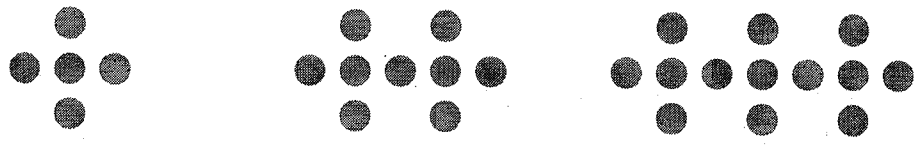
(c) Is Kyle right?
You must give a reason for your answer.

No. There will be 7 white tiles in pattern 6.

(1)

(Total for Question 11 is 4 marks)

12 Here is a sequence of patterns made from grey counters.

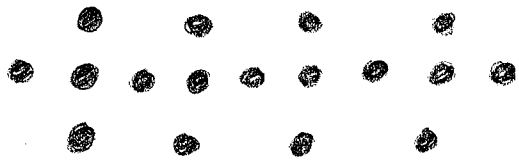


pattern number 1 — 5

pattern number 2 — 9

pattern number 3 — 13

(a) In the space below, draw pattern number 4.



(1)

(b) Work out the total number of counters to make pattern number 10.

	5	9	13	17
4n	4	8	12	16

$4n + 1$

$4(10) + 1 = 41$

41

(2)

(Total for Question 12 is 3 marks)

---

13 Here are the first five terms of a sequence.

31 27 23 19 15

(a) Find the first negative term in the sequence.

11 7 3 −1

−1

(2)

(b) Is –30 a term in this sequence?
Give a reason for your answer.

No. All terms in the sequence are odd

(1)

(Total for Question 13 is 3 marks)

14 Here are the first 5 terms of an arithmetic sequence.

-3 1 5 9 13

(a) Find an expression, in terms of n, for the nth term of this sequence.

$4n$ 4 8 12 16 20

$4n - 7$

(2)

The nth term of a different arithmetic sequence is $2n - 3$

(b) Is 101 a term in this sequence?
Show how you get your answer.

$2n - 3 = 101$
$2n = 104$
$n = 52$

Yes, it is the 52nd term.

(2)

(Total for Question 14 is 4 marks)

---

15 Here are the first 5 terms of a sequence.

9 14 19 24 29

Find an expression, in terms of n, for the nth term of this sequence.

$5n$ 5 10 15 20 25

$5n + 4$

(Total for Question 15 is 2 marks)

---

16 Here are the first 5 terms of a sequence.

25 22 19 16 13

Find an expression, in terms of n, for the nth term of this sequence.

$-3n$ -3 -6 -9 -12 -15

$-3n + 28$

(Total for Question 16 is 2 marks)

17 Here are the first four terms of an arithmetic sequence.

4 11 18 25

Write down an expression, in terms of n, for the nth term of the sequence.

$7n$ 7 14 21 28

$7n - 3$

(Total for Question 17 is 2 marks)

---

18 Here are the first four terms of an arithmetic sequence.

35 31 27 23

Write down an expression, in terms of n, for the nth term of the sequence.

$-4n$ -4 -8 -12 -16

$-4n + 39$

(Total for Question 18 is 2 marks)

---

19 Here are the first five terms of an arithmetic sequence.

21 27 33 39 45

Write down an expression, in terms of n, for the nth term of the sequence.

$6n$ 6 12 18 24 30

$6n + 15$

(Total for Question 19 is 2 marks)

---

20 Here are the first five terms of an arithmetic sequence.

2 7 12 17 22

Write down an expression, in terms of n, for the nth term of the sequence.

$5n$ 5 10 15 20 25

$5n - 3$

(Total for Question 20 is 2 marks)